Ch 5.2: The Bootstrap

Lecture 12 - CMSE 381

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Announcements

Last time:

- k-fold CV

Announcements:

- Fifth homework due Friday! Covers:
  - Fri 2/11 Lab
  - Mon 2/14 Lecture
  - Weds 2/16 Lecture
- Office hours normal this week.
- Next week office hours
  - Dr Munch's OH canceled.
  - Emily's OH are still at normal times
  - Send slack messages!

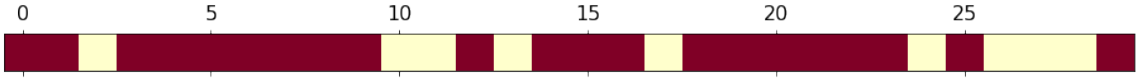
Covered in this lecture

- Bootstrap

Section 1

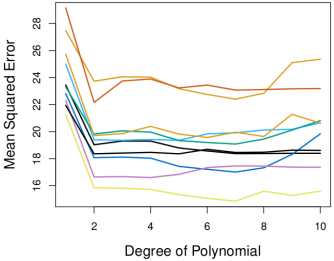
Last time

Validation set approach

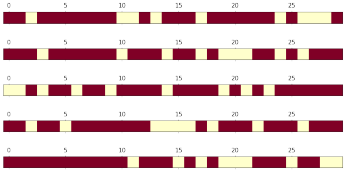


- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

Problems



Ex. Predict `mpg` using `horsepower`



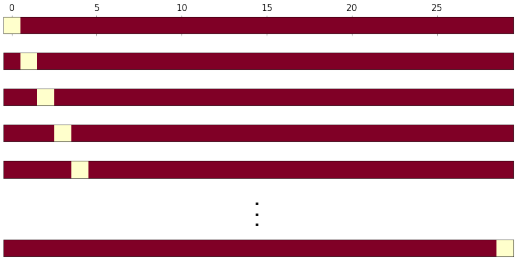
- Highly variable results, no consensus about the error
- Tends to overestimate test error rate

Leave One Out CV (LOOCV)

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

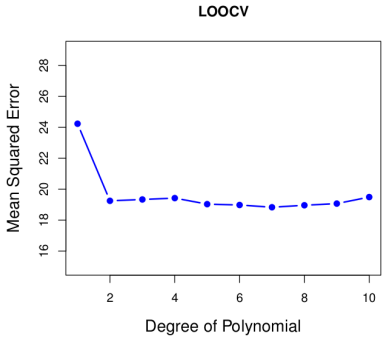
- Rinse and repeat



Return the score:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Pros and Cons



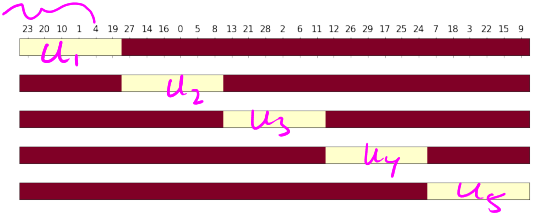
No randomness!

- No variance
- Higher computation cost

k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

$k = 5$



- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$

  # pts in fold
- Rinse and repeat

Return

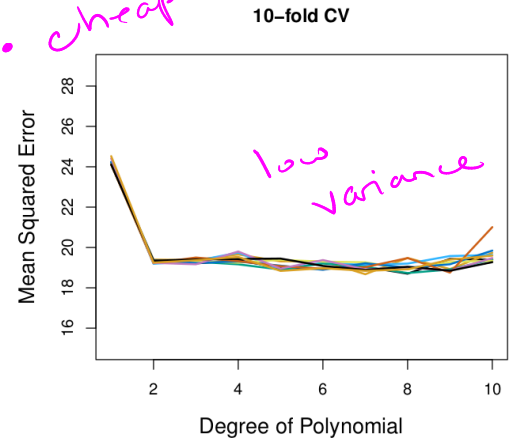
$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

$k = n \Rightarrow$ LOOCV

Pros and Cons

Pros:

- cheaper computation



Cons:

has randomness but less variability

Bias-Variance Tradeoff

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \underbrace{\left[\mathrm{Bias}(\hat{f}(x_0))\right]^2} + \mathrm{Var}(\varepsilon)$$

Higher Bias

- Validation set overestimates test error b/c used small subset of data
- k-fold gives medium level of bias b/c training set has approximately $(k-1)n/k$ observations
- LOOCV gives approximately unbiased estimate since uses almost all data every time

Lower Bias

Higher Variance

- LOOCV: avg the outputs of n fitted models with high correlation
- k-fold: k fitted models somewhat less correlated with each other

Lower Variance

Usually use $k = 5$ or $k = 10$

Section 2

The Bootstrap

The Idea

The goal: quantify the uncertainty associated with a given estimator or statistical learning method.

The bootstrap

Boring b/c we can do it w/ older tools

- Can be used to estimate standard error of
  - linear reg. coeff
  - ~~~~
- The power of the bootstrap is that it can be applied to many statistical methods

The boostrap idiom

* Use the data we have to create samples *

"Pull yourself up by your bootstraps."

- Originally used as a saying meaning an impossible task, used sarcastically
- Now often used to imply that socioeconomic advancement is something everyone should be able to do

Booting a computer

Today's class: Bootstrap on a simple modeling problem

- We wish to invest a fixed sum of money in two financial assets that yield returns of X and Y, respectively, where X and Y are random quantities.
- We will invest a fraction α of our money in X, and will invest the remaining $1 - \alpha$ in Y.
- Since there is variability associated with the returns on these two assets, we wish to choose α to minimize the total risk, or variance, of our investment.

$\alpha \in [0,1]$

Goal: Minimize

$$\mathrm{Var}(\alpha X + (1-\alpha) Y)$$

One can show.....

...that $\mathrm{Var}(\alpha X + (1-\alpha)Y)$
is minimzed by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

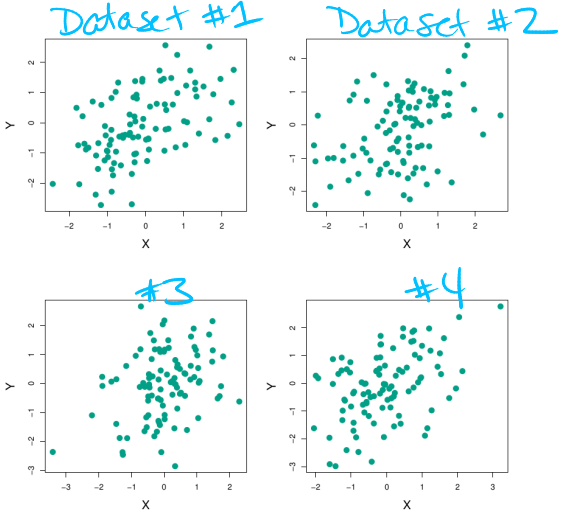
where

- $\sigma_X^2 = \mathrm{Var}(X)$
- $\sigma_Y^2 = \mathrm{Var}(Y)$
- $\sigma_{XY} = \mathrm{Cov}(X, Y)$

Get an estimate:

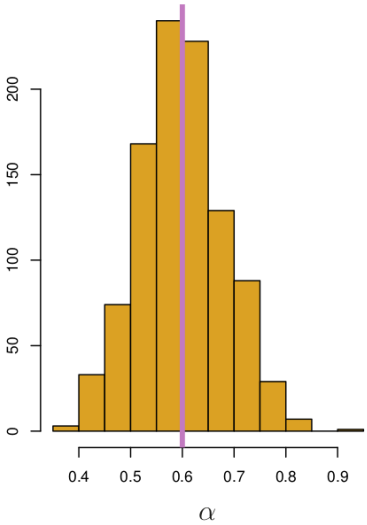
$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

Simulated data



- Simulate data using
  - $\sigma_X^2 = 1$
  - $\sigma_Y^2 = 1.25$
  - $\sigma_{XY} = 0.5$
  - Implies $\alpha = 0.6$
- In each panel: Simulate 100 pairs of returns for investments
- Predict σ_X^2, σ_Y^2, and σ_{XY}
- $\hat{\alpha}$ prediction by panel:
  0.576 0.543
  0.657 0.651

Resimulate: Rinse and repeat



- Simulate 100 data points 1,000 times
- Left: Histogram of predictions for α
- Pink line: True value for α
- Mean over simulated values:
  $0.5996 = \bar{\alpha} = \boxed{\frac{1}{1000}\sum \hat{\alpha}_r}$
- St dev: $0.083 = \sqrt{\frac{1}{1000-1}\sum_r (\hat{\alpha}_r - \bar{\alpha})}$

So what's the problem?

I can't simulate my data!

The plan

Use a given dataset to create simulations

The solution: Sample the data with replacement

Obs	X	Y
1	4.3	2.4
2	2.1	1.1
3	5.3	2.8

Original Data (Z)

Z^{*1}

Obs	X	Y
3	5.3	2.8
1	4.3	2.4
3	5.3	2.8

$\hat{\alpha}^{*1}$

Z^{*2}

Obs	X	Y
2	2.1	1.1
3	5.3	2.8
1	4.3	2.4

$\hat{\alpha}^{*2}$

Z^{*B}

Obs	X	Y
2	2.1	1.1
2	2.1	1.1
1	4.3	2.4

$\hat{\alpha}^{*B}$

Get B sampled data sets

- Data pts can show up repeatedly
- Starting data had n pts
  Samples each have n
- Unlike validation
  - have repeats
  - could have data pt never seen

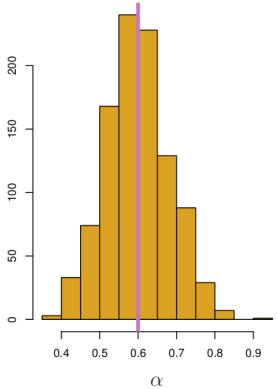
Computation of error

- Repeate procedure B times:
- Get B bootstrap data sets, $Z^{*1}, Z^{*2}, \ldots, Z^{*B}$
- Get B bootstrap estimates $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \ldots, \hat{\alpha}^{*B}$

Get standard error estimate:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \underbrace{\frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}}_{\text{average } \hat{\alpha} \text{ estimate}}\right)^2}$$

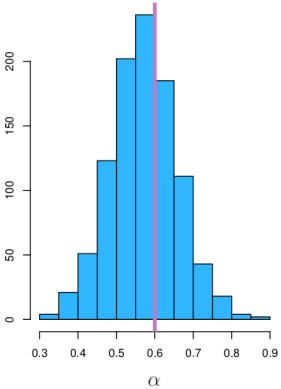
Back to the example



Resample version

Predicted $SE(\hat{\alpha}) = 0.083$

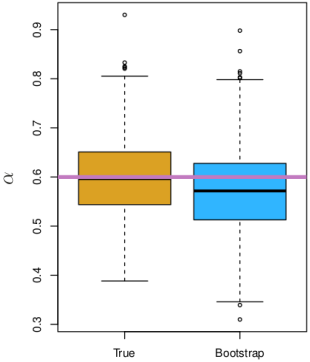
ground truth



Bootstrap version

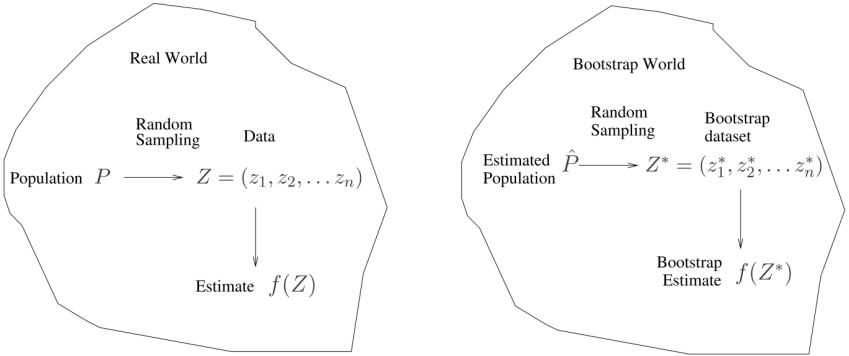
Predicted $SE(\hat{\alpha}) = 0.087$

guess



Spread of both is similar

A general picture for the bootstrap



TL;DR

- Start with data set of n points
- Sample n points with replacement to get data set Z^{*1}
- Use this to estimate whatever parameter we want $\hat{T}^{*1}$

- Repeat B times to get estimates $\hat{T}^{*1}, \cdots, \hat{T}^{*B}$
- Estimate standard error of our T estimate by

$$SE_B(\hat{T}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{T}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{T}^{*r'}\right)^2}$$

Boostrap vs Cross-Validation

Bootstrap:

Resample w/ replacement

Same # pts per data set as n (original set size)

Randomness: B times, replacement

Might not see all pts

Goal: Establish emperical distribution

CV:

w/o replacement

Subset, so <n pts

LOOCV no random.

k-fold random from sorting

Sees all pts

Measuring performance of model

Section 3

Looking ahead: 6.1 Linear Model Selection and Regularization

Goals of fitting a given model

Goal minimize RSS = $\sum e_i^2 = \sum (y_i - \hat{y}_i)^2$

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation.

$\hat{\beta}_{17} = 0.0003$

Prediction accuracy

If true relationship is linear

least squares estimates have low bias

If $n \gg p$ have low variance

If n not much larger than p high variability

If $n < p$, can't do Least squares

Model Interpretability

Often, many variables (b.g p), but many not assoc. w/ response

"Use fewer variables"

have coef. of those variables be close to 0

Goal of next chapter

Drop RSS minimization / least squares estimate

replace/modify "something else"

- Subset selection
  identify a subset of variables that are important then least squares on those
- Shrinkage
  force some, many estimated coeff to be closer to 0.
- Dimension reduction
  project data into lower dim. space MLR

after midterm

Next time

500 pts

- Fri: Resampling
  - Lab on ~~Classifiers~~
  - HW #5 due. Covers:
    - 2/11: Classification Lab
    - 2/14: 5.1.3-5.1.5 k-fold CV; CV for classification
    - 2/16: 5.2 Bootstrap

Next week:

- Mon 2/21: Guest Lecture, Dr. Yuying Xie
- Weds 2/23: Online only lecture, 6.1 Subset Selection
- Fri 2/25: Online only lab, 6.5.1 Subset Selection Methods

The week after:

- Mon 2/28: 6.2 Shrinkage
- Weds 3/2: Finish shrinkage + Review session
  - Bring your questions!
- Fri 3/4: Midterm
  - In person
  - Difficulty on par with homeworks
  - Cheat sheet: One 8.5x11 paper.
  - Handwritten
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I was right, this one was SUPER short. Could definitely combine this guy with the k-folds day next time